

LASER PHYSICS:

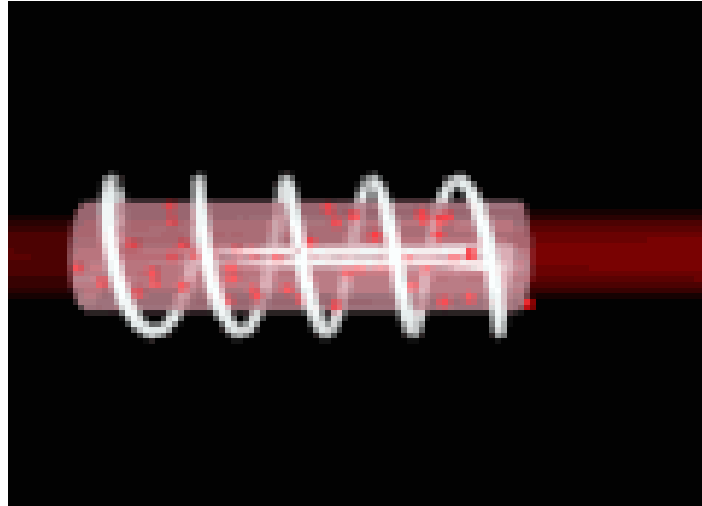
- Introduction to lasers
- Conditions for laser action
- Requisites of a Laser system Principle
- Construction and working of Nd-YAG
- Construction and working of Semiconductor Laser
- Application

Lasers in Defense (Laser range finder),

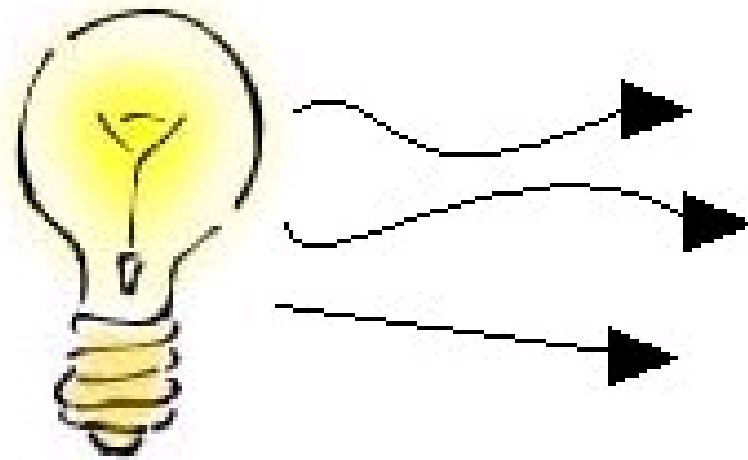
Engineering (Data storage) and

Applications of Lasers in medicine

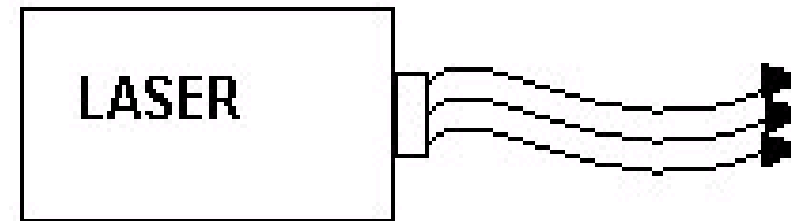
LASERS



Incandescent vs. Laser Light



- 1. Many wavelengths***
- 2. Multidirectional***
- 3. Incoherent***
- 4. Less Intensity***



- 1. Monochromatic**
- 2. Directional**
- 3. Coherent**
- 4. High Intensity**

What is Laser?

- **Light Amplification by Stimulated Emission of Radiation**
- A device produces a coherent beam of optical radiation by stimulating electronic, ionic, or molecular transitions to higher energy levels.
- When they return to lower energy levels by stimulated emission, they emit energy.

History of the LASER

- Invented in 1958 by Charles Townes (Nobel prize in Physics 1964) and Arthur Schawlow of Bell Laboratories.
- Was based on Einstein's idea of the “particle wave duality” of light, more than 30 years earlier.
- Originally called MASER (m = “microwave”)



Properties of Laser

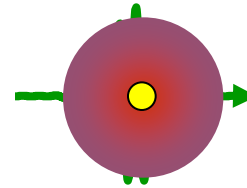
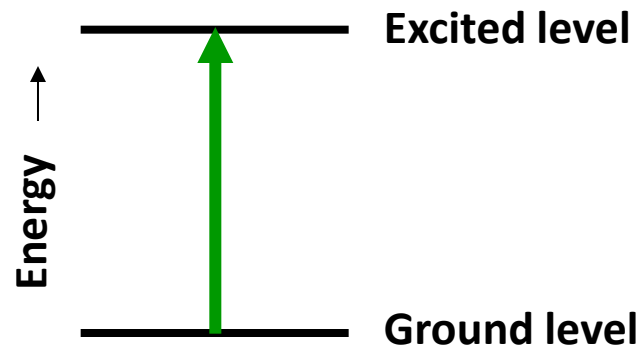
- The light emitted from a laser is **monochromatic**, that is, it is of one color/wavelength. In contrast, ordinary white light is a combination of many colors (or wavelengths) of light.
- Lasers emit light that is highly **directional**, that is, laser light is emitted as a relatively narrow beam in a specific direction. Ordinary light, such as from a light bulb, is emitted in many directions away from the source.
- The light from a laser is said to be **coherent**, which means that the wavelengths of the laser light are in phase in space and time. Ordinary light can be a mixture of many wavelengths.

These three properties of laser light are what can make it more hazardous than ordinary light. Laser light can deposit a lot of energy within a small area.

Basic principle of LASER

- Stimulated Absorption or Induced Absorption or Excitation
- Life time
- Spontaneous Emission
- Meta stable states
- Stimulated Emission
- Lasing action

Stimulated Absorption or Induced Absorption or Excitation

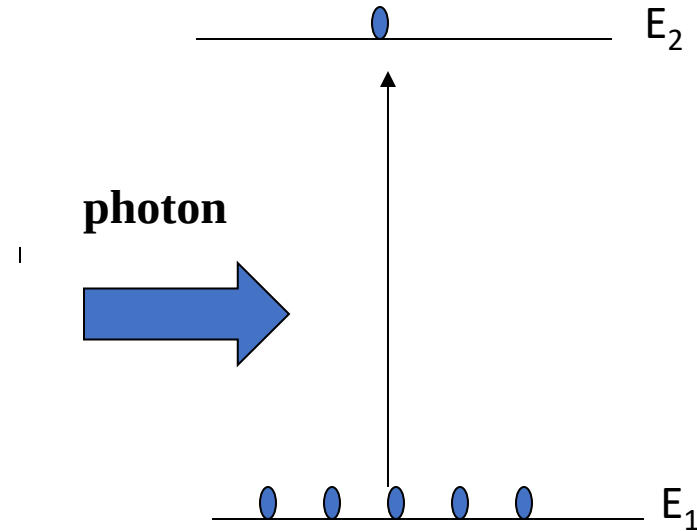


*This is, of course,
absorption.*

Atoms and molecules can also **absorb** photons, making a transition from a lower level to a more excited one.

Absorption

- Let us consider two energy level having energy E_1 & E_2 respectively.
- The atom will remain in ground state unless some external stimulant is applied to it.
- When an EM wave i.e photon of particular frequency fall on it, there is finite probability that atom will jump form energy state E_1 to E_2 .

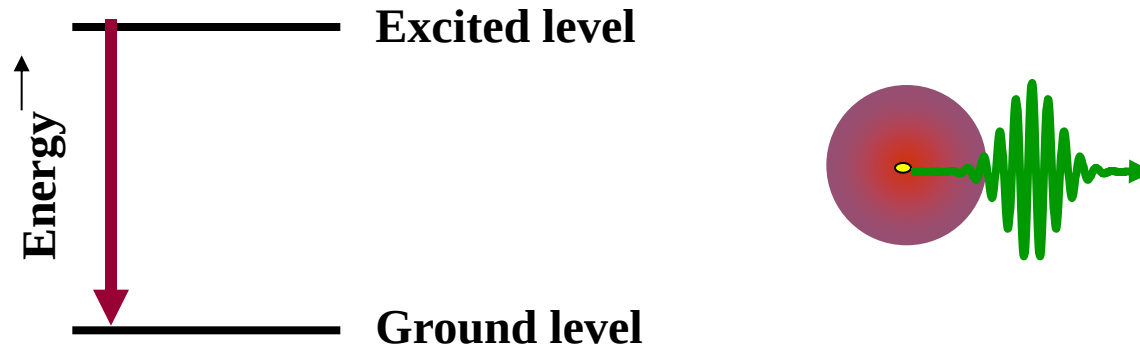


Life - Time

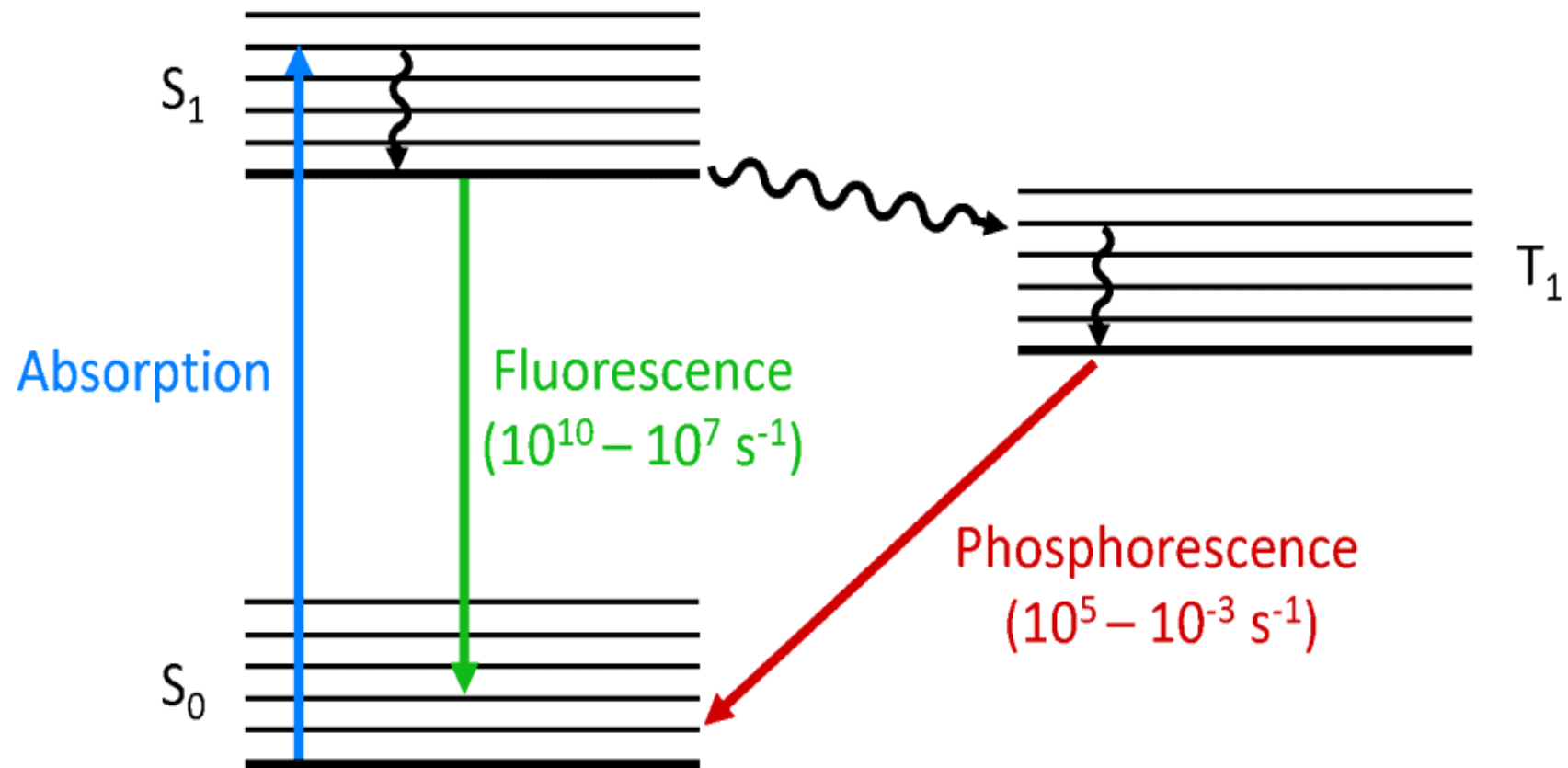
**The time for which the atom can remain in the
excited state is known as life- time
(10^{-9} sec)**

Spontaneous Emission

When the transition takes place from excited state to ground state without any external force.

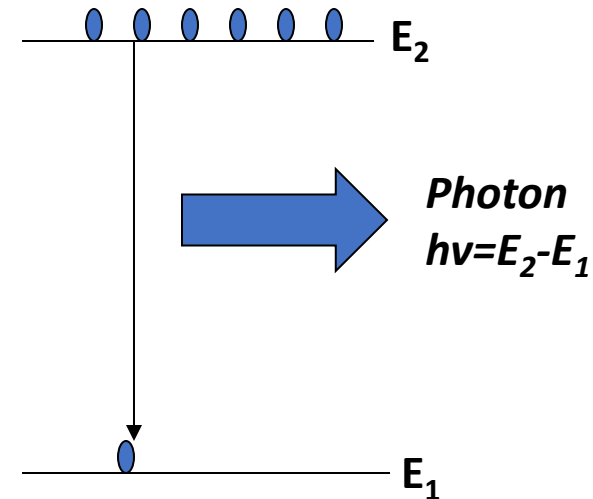


Molecules typically remain excited for no longer than a few nanoseconds. This is often also called **fluorescence** or, when it takes longer, **phosphorescence**.



Characteristics of Spontaneous emission

- Consider an atom in higher state (E_2).
- It can decay to lower energy level by emitting photon.
- Emitted photon have energy $h\nu = E_2 - E_1$.
- Life time of excited state is 10^{-9}sec .



Meta Stable State

The excited state in which the atoms can remain more than the life time.

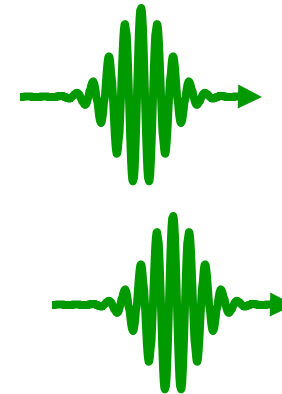
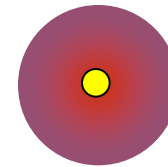
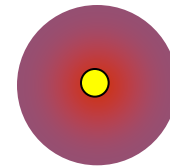
Stimulated Emission

When the transition takes place from excited state to ground state with an external force

**Stimulated
emission**



Photon



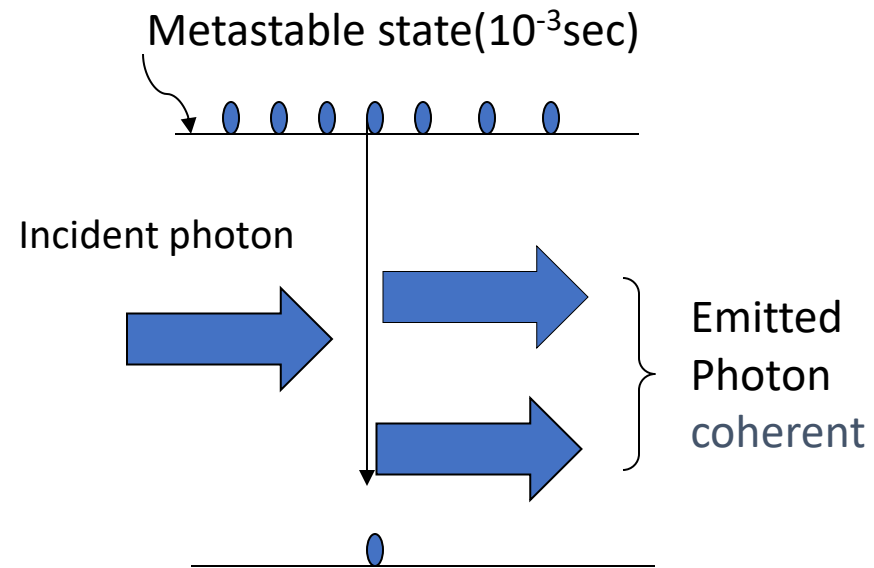
**Two Photons
having same
energy**

Characteristics of Stimulated emission

- There are meta-stable states i.e. transition from this state is not allowed act to selection rule.
- There life time is 10^{-3} sec.



- Atom in this state can't jump to lower state at there own.
- When an photon of suitable freq arrive it make the atom in metastable unstable.
- The emitted photon is in coherence with incident photon.

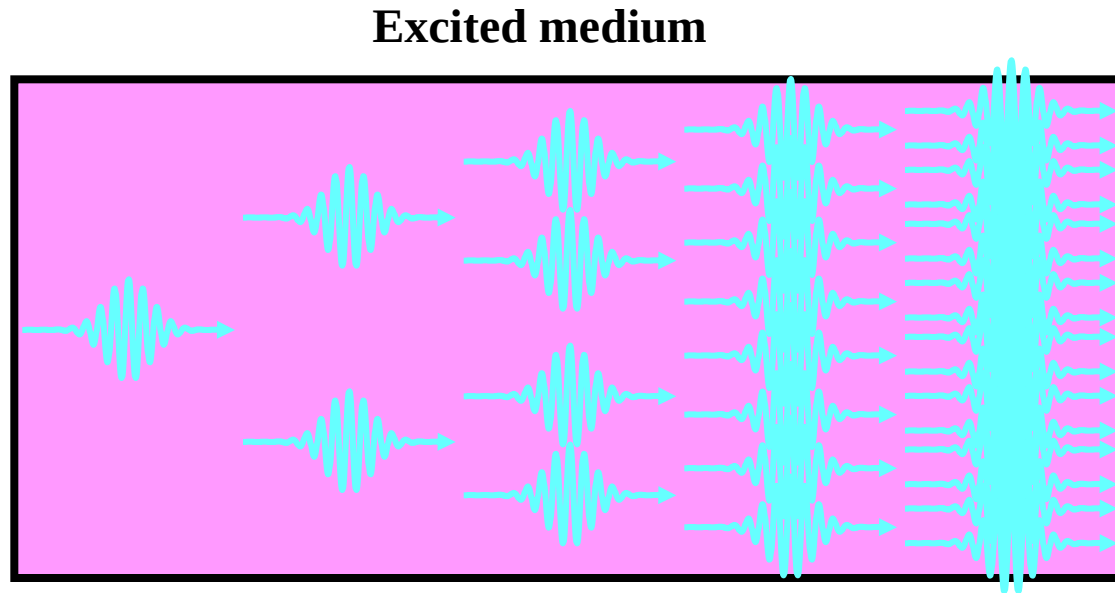


Population Inversion

- A state in which a substance has been energized, or excited to specific energy levels.
- Generally more number of the atoms present in the ground state than the excited state.
- *The process of making more number of atoms in the excited state than ground state is known as **population inversion***

Lasing Action

If a medium has many excited molecules, one photon can become many.



This is the **essence of the laser**. The factor by which an input beam is amplified by a medium is called the **gain** and is represented by G .

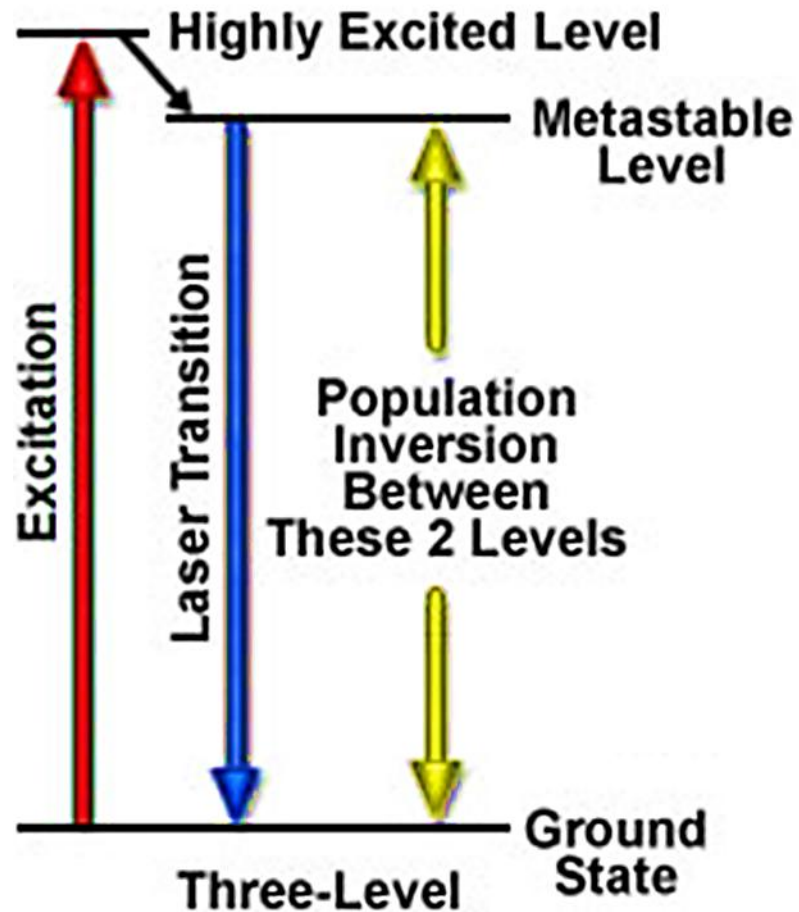
Two-level Laser System

Un-imaginable

*as absorption and stimulated processes
neutralize one another.*

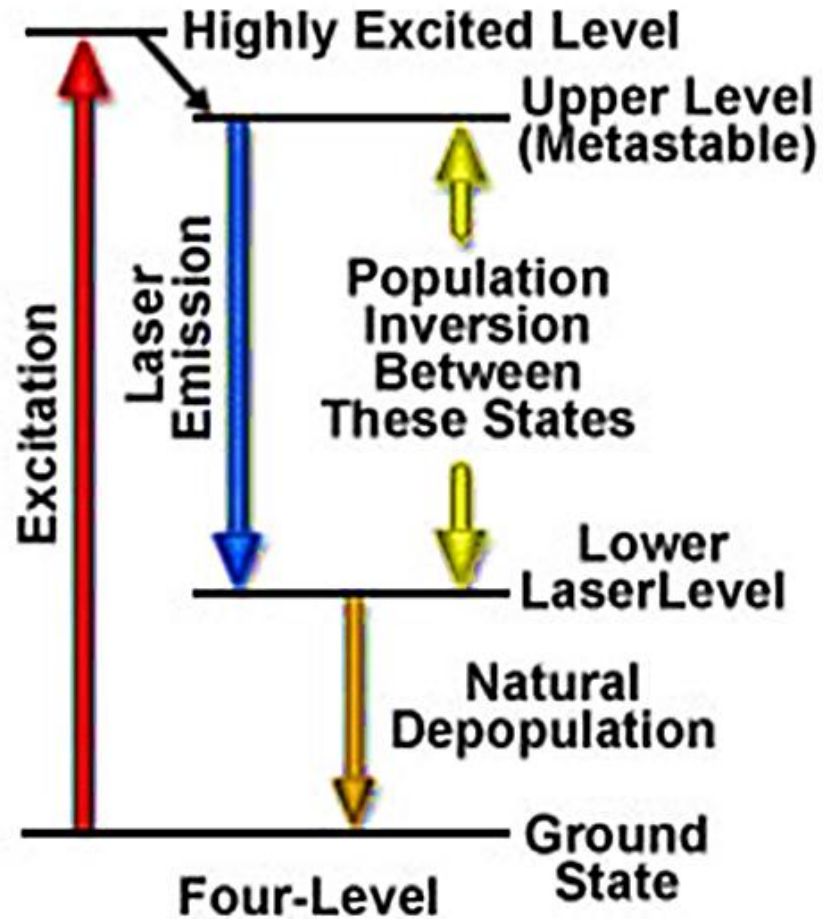
The material becomes transparent.

Three-level Laser System



- Initially excited to a short-lived high-energy state .
- Then quickly decay to the intermediate meta-stable level.
- Population inversion is created between lower **ground state** and a higher-energy **meta-stable state**.

Four-level Laser System



- Laser transition takes place between the **third and second** excited states.
- Rapid **depopulation** of the lower laser level.

Requisites or condition for LASER system or action

1. Active medium
2. Pumping and coherent emission
3. Population inversion
4. Resonant activity

1. Active medium:

A material medium in which the population inversion and lasing action can be achieved

- Examples :
- (i) Gas laser (He-Ne laser and CO₂ laser)
 - (ii) Solid state laser (Ruby laser and NdYAG laser)
 - (iii) Semi conductor laser (GaAs laser)
 - (iv) Liquid laser (Dye laser)

2. Pumping and coherent emission:

To raise the atoms from lower to higher energy state by supplying sufficient energy is called optical pumping

- Examples :
- (i) Optical pumping (Ruby laser and NdYAG laser)
 - (ii) Electrical discharge (He-Ne laser and CO₂ laser)
 - (iii) Direct conversion (LED and semiconductor laser)
 - (iv) Chemical reaction (Dye laser)

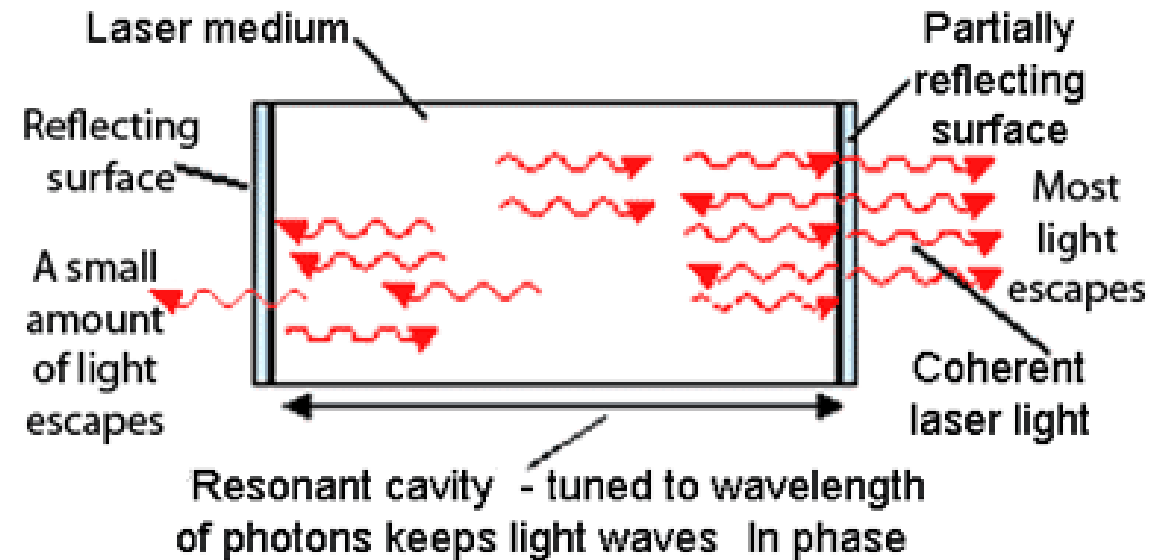
3. Population inversion:

Under normal condition number of atoms in the ground state will be more than the number of atoms in the excited state. For laser action the **number of atoms in the excited state should be more than the number of atoms in the ground state** this phenomenon is called population inversion

Example: Xenon flash tube

4. Resonant activity:

In order to generate a coherent photon it is necessary that stimulated photons are to be made to pass through the active medium number of times.



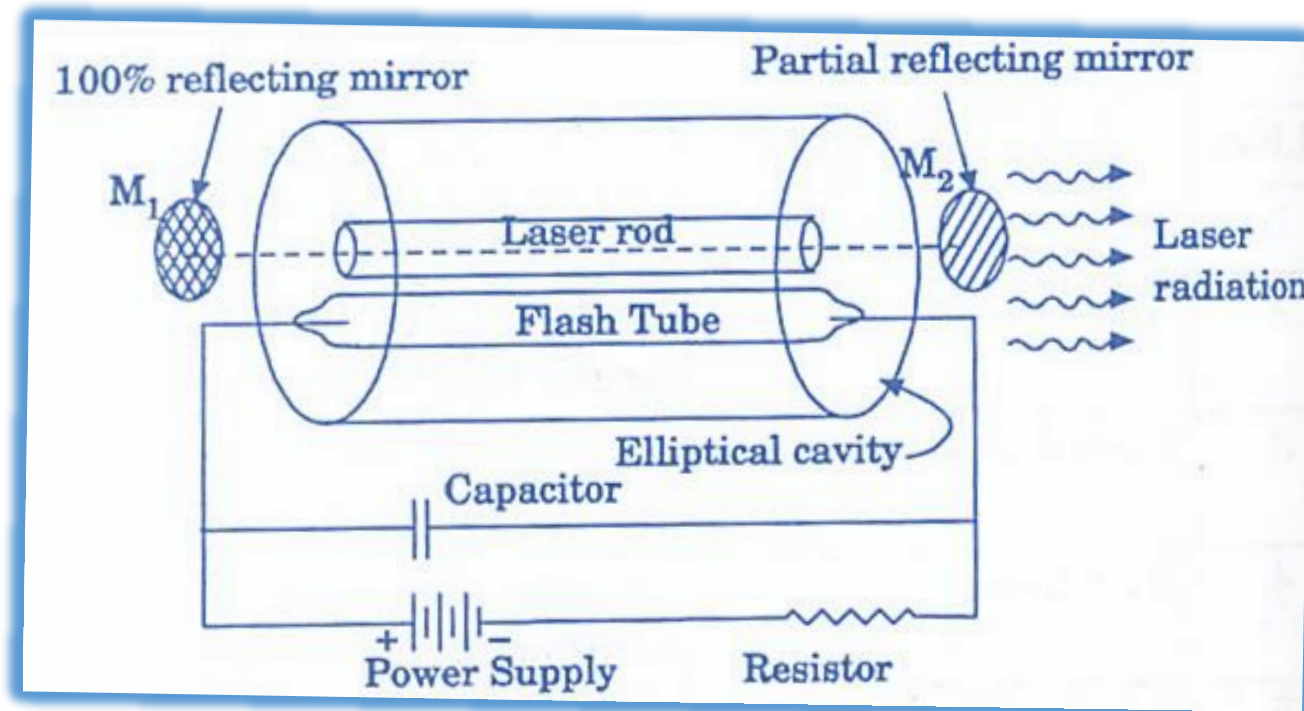
Nd: YAG laser:

Nd: YAG laser is a neodymium based laser. Nd stands for Neodymium (rare earth element) and YAG stands for **Yttrium Aluminum Garnet** ($\text{Y}_3\text{Al}_5\text{O}_{12}$). It is a four level solid state laser.

- **Principle**
- **Construction**
- **Working, Characteristics**
- **Advantages**
- **Disadvantages**
- **Applications**

Principle

The active medium Nd: YAG rod is optically pumped by **Krypton flash tubes**. The Neodymium ions (Nd^{3+}) are raised to excited levels. During the transition from meta stable state to ground state, a laser beam of wavelength $1.064\text{ }\mu\text{m}$ is emitted.

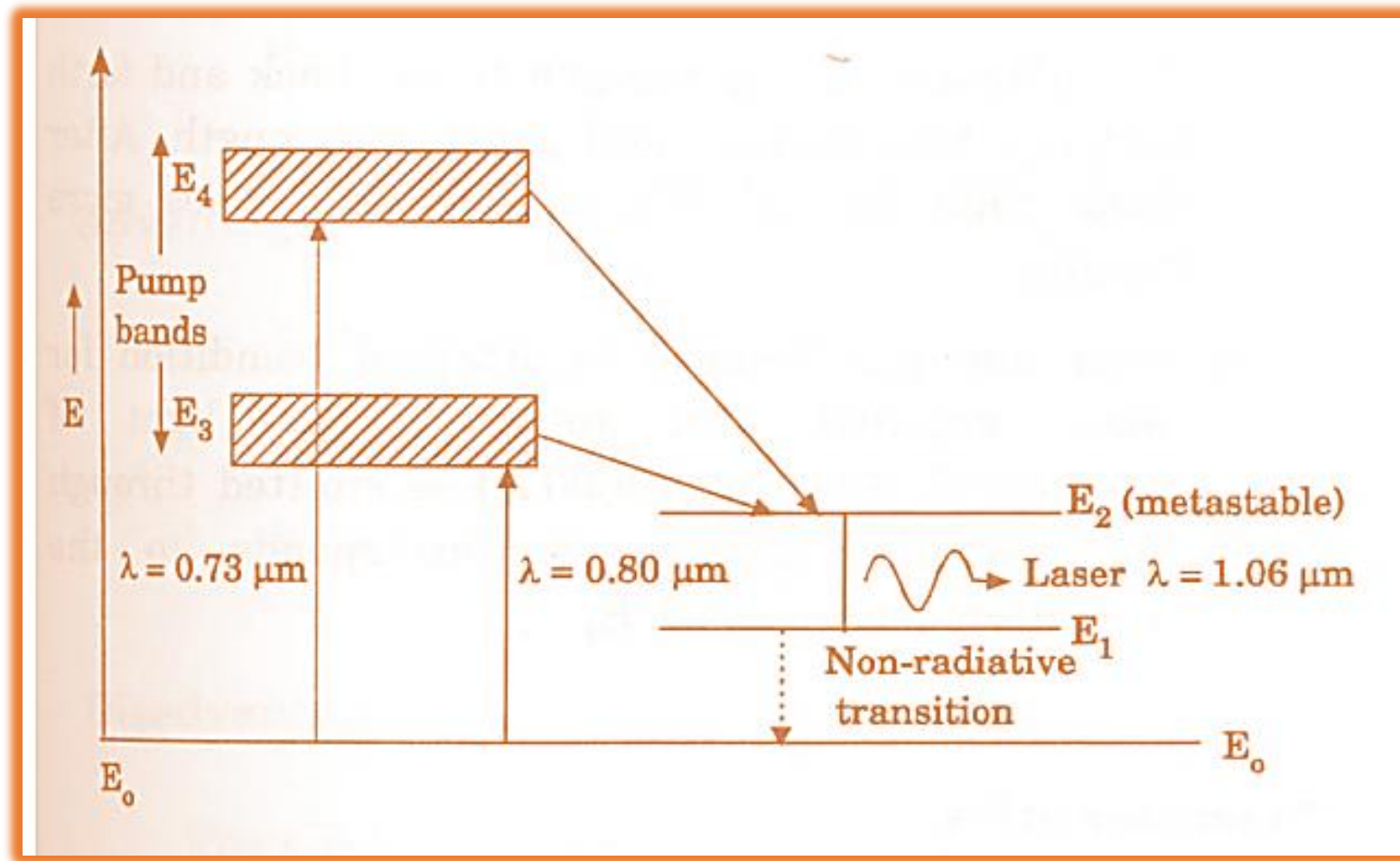


Construction

- The construction of Nd: YAG laser is as shown in the Figure. **A small amount of Yttrium ions (Y^{3+}) is replaced by Neodymium (Nd^{3+}) in the active element of Nd: YAG crystal.**
- This active element is cut into a **cylindrical rod**. The ends of the cylindrical rod are highly **polished** and they are made **optically flat and parallel**.
- This cylindrical rod (**laser rod**) and a pumping source (**flash tube**) are placed inside a highly (reflecting) **elliptical reflector cavity**.
- The optical resonator is formed by using **two external reflecting mirrors**. One mirror (M1) is 100% reflecting while the other mirror (M2) is partially reflecting.

Working

Figure shows the energy level diagram for Nd: YAG laser. These energy levels are those of **Neodymium (Nd^{3+}) ions**.



- When **the krypton flash lamp is switched on**, by the absorption of light radiation of **wavelength $0.73\mu\text{m}$ and $0.8\mu\text{m}$** , the Neodymium (Nd^{3+}) atoms are raised **from ground level E_0 to upper levels E_3 and E_4 (Pump bands)**.
- The Neodymium ions atoms make a transition from these energy **levels E_2 by non-radiative transition. E_2 is a metastable state.**
- The Neodymium ions are **collected in the level E_2 and the population inversion is achieved between E_2 and E_1 .**
- An ion makes a spontaneous transition from E_2 to E_1 , emitting a photon of energy $h\nu$. This emitted photon **will trigger a chain of stimulated photons between E_2 and E_1 .**
- The **photons thus generated travel back and forth between two mirrors and grow in strength.** After some time, the photon number multiplies more rapidly.
- After enough strength is attained (**condition for laser being satisfied**).
- An **intense laser light of wavelength $1.06\mu\text{m}$** is emitted through the partial reflector. It corresponds to the transition **from E_2 to E_1 .**

Characteristics:

1. **Type:** It is a **four level** solid state laser.
2. **Active medium:** The active medium is **Nd: YAG laser**.
3. **Pumping method:** **Optical pumping** is employed for pumping action.
4. **Pumping source:** **Xenon or Krypton flash** tube is used as pumping source.
5. **Optical resonator:** Two ends of Nd: YAG rod is **polished with silver (one end is fully silvered and the other is partially silvered)** are used as optical resonator.
6. **Power output:** The power output is approximately **70 watt**.
7. **Nature of output:** The nature of output is **pulsed or continuous beam of light**.
8. **Wavelength of the output:** The wavelength of the output beam is **1.06 μ m (infra-red)**

Advantages:

1. It has **high energy** output.
2. It has very **high repetition rate** operation
3. It is much **easy to achieve population inversion**.

Disadvantages:

The **electron energy level** structure of Nd^{3+} in YAG is **complicated**.

Applications:

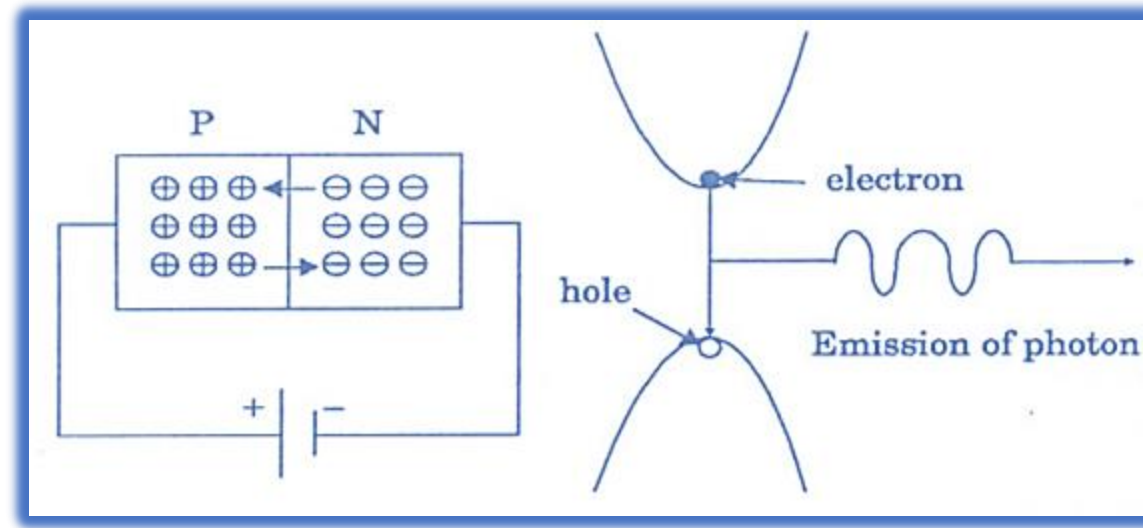
1. It finds many applications in **range finders** and illuminators.
2. It is widely used in **engineering applications** such as resistor, trimming scribing, micro machining operations as well as welding, drilling etc.
3. It finds many **medical applications** such as **endoscopy, urology, neurosurgery, ENT, gynecology, dermatology, dental surgery** and general surgery.

Semiconductor Diode laser

It is specifically fabricated **p-n junction diode**. This diode emits laser light when it is forward biased.

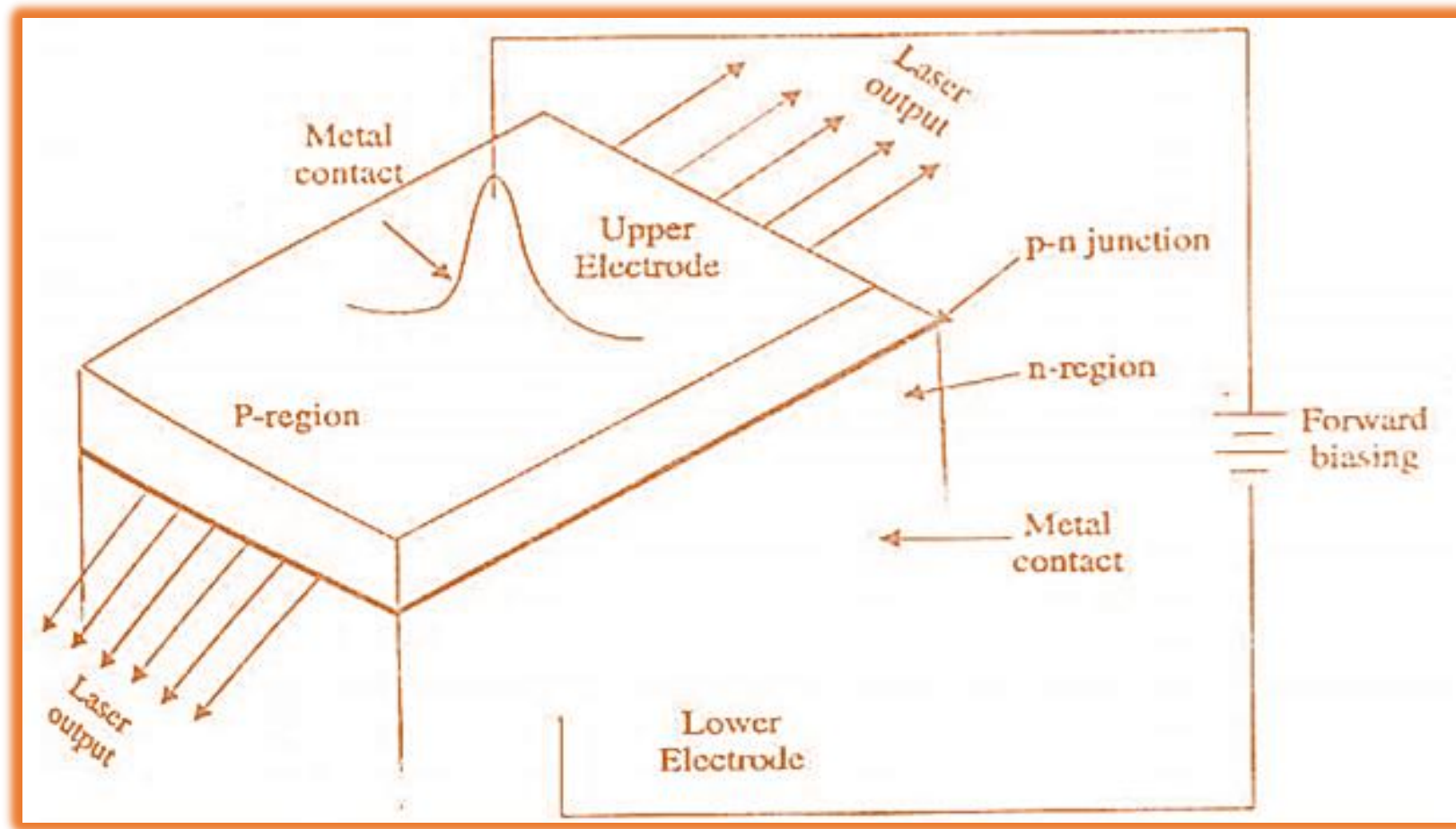
- **Principle**
- **Construction**
- **Working, Characteristics**
- **Advantages**
- **Disadvantages**
- **Applications**

Principle:



- When a **p-n junction diode** is **forward biased**, the electrons from n– region and the holes from the p– region cross the junction and **recombine with each other**.
- During the **recombination process**, the light radiation (photons) is released from a **certain specified direct band gap semiconductors like Ga-As**. This light radiation is known as **recombination radiation**.
- The photon **emitted during recombination stimulates other electrons and holes to recombine**. As a result, stimulated emission takes place which produces laser.

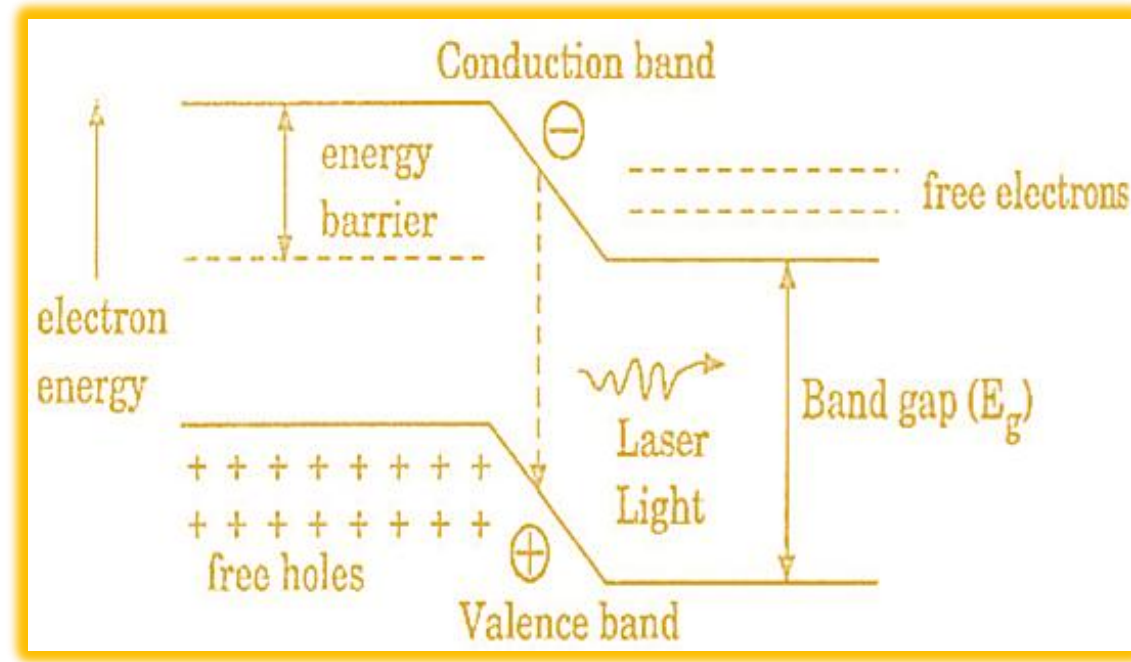
Construction:



- Figure shows the basic construction of semiconductor laser. The active medium is a p-n junction diode made from the **single crystal of gallium arsenide**.
- This crystal is cut in the form of a platter having **thickness of $0.5\mu\text{m}$** .
- The platelet consists of **two parts having** an electron conductivity (n-type) and hole conductivity (p-type).
- The **photon emission is stimulated in a very thin layer of PN junction** (in order of few microns). The electrical voltage is applied to the crystal through the electrode fixed on the upper surface.
- The **end faces of the junction diode** are well polished and parallel to each other.
- They act as **an optical resonator** through which the **emitted light comes out**.

Working:

Figure shows the energy level diagram of semiconductor laser.



- When the **PN junction is forward biased with large applied voltage**, the electrons and holes are **injected into junction region** in considerable **concentration**.
- The region around the junction **contains a large amount of electrons** in the **conduction band** and a large amount **of holes in the valence band**.

- If the **population density is high**, a **condition of population inversion is achieved**. The **electrons and holes recombine with each other** and this recombination's produce **radiation in the form of light**.
- When the **forward-biased voltage is increased**, more and more light photons are emitted and the light production **instantly becomes stronger**. These **photons will trigger a chain of stimulated recombination** resulting in the release of photons in phase.
- **The photons moving at the plane of the junction travels** back and forth by reflection between two sides placed parallel and opposite to each other and **grow in strength**.
- After **gaining enough strength**, it gives out the **laser beam of wavelength 8400 A⁰** . The wavelength of laser light is given by

$$E_g = h\nu = h \frac{c}{\lambda}$$

$$\lambda = \frac{hc}{E_g}$$

Where E_g is the band gap energy in joule.

Characteristics:

1. **Type:** It is a **solid state semiconductor laser**.
2. **Active medium:** A PN junction diode made from **single crystal of gallium arsenide** is used as an active medium.
3. **Pumping method:** The **direct conversion method** is used for pumping action
4. **Power output:** The power output **from this laser is 1mW**.
5. **Nature of output:** The nature of output is **continuous wave or pulsed output**.
6. **Wavelength of Output:** gallium arsenide laser **gives infrared radiation in the wavelength 8300 to 8500 Å**.

Advantages:

1. It is very **small in dimension**. The arrangement is simple and compact.
2. It exhibits **high efficiency**.
3. The laser output can be **easily increased by** controlling the junction current
4. It is operated with **lesser power than** ruby and CO₂ laser.
5. It requires very **little auxiliary** equipment.
6. It can have a **continuous wave output or pulsed** output.

Disadvantages:

1. It is difficult to **control the mode pattern and mode structure** of laser.
2. The output is usually from **5 degree to 15 degree** i.e., laser beam has large divergence.
3. The purity and monochromaticity are less than other types of laser.
4. Threshold **current density** is very large (400A/mm^2).
5. It **has poor coherence and poor stability**.

Application:

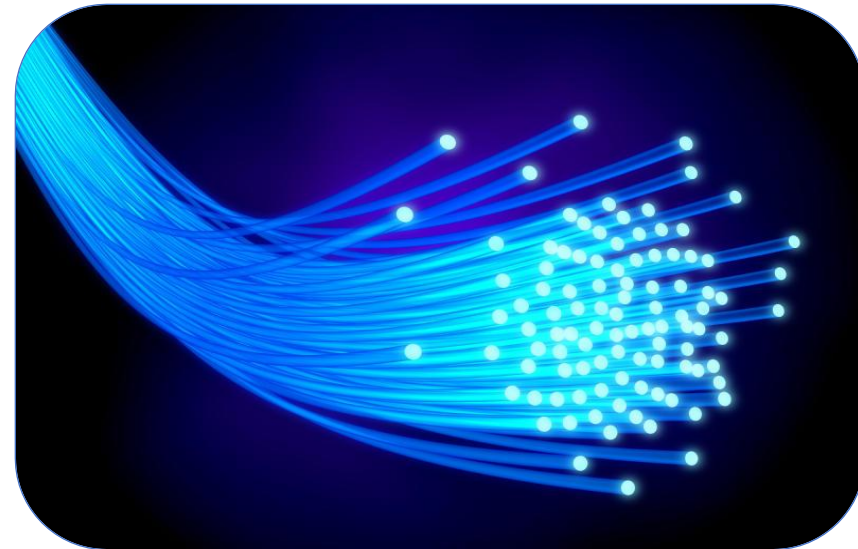
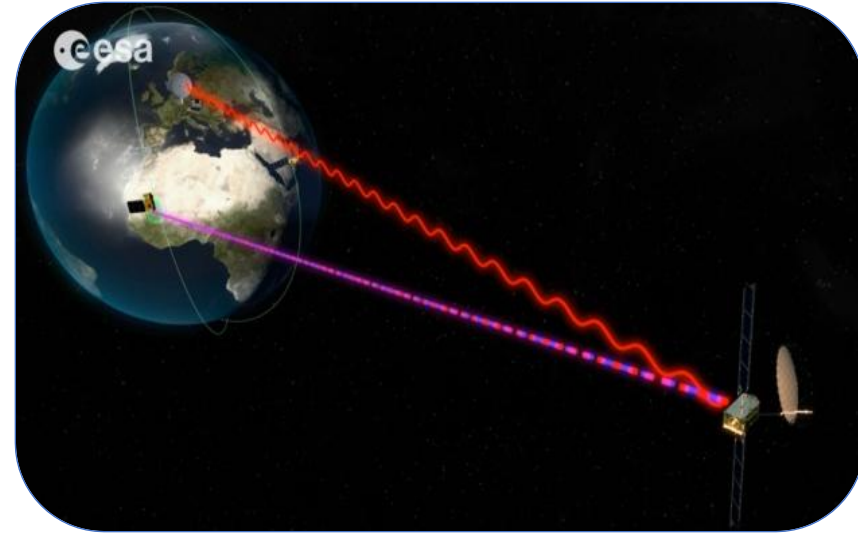
1. It is widely used in **fiber optic communication**
2. It is used to heal the wounds by **infrared radiation**
3. It is also used as a **pain killer**
4. It is used in laser printers and **CD writing and reading.**

Applications of Lasers

Lasers in Communication:

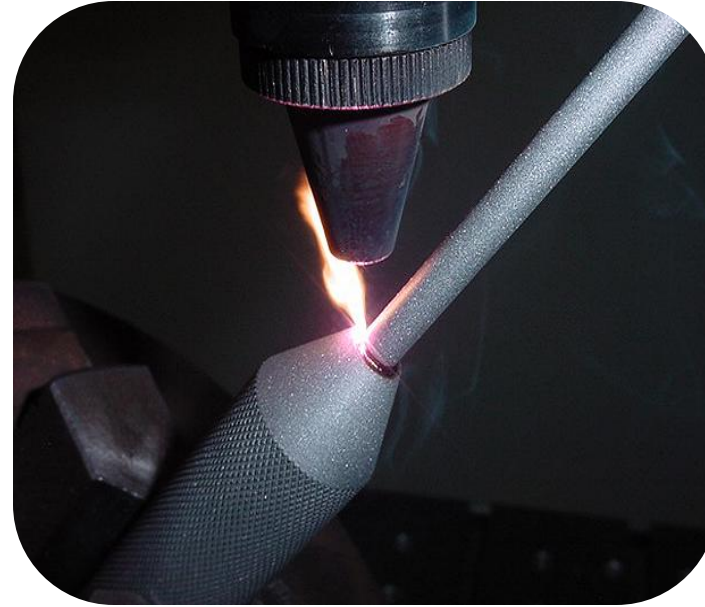
Lasers are used in Optical fibre communication as light source to transmit audio, video signals and data to long distances without attenuation and distortion.

- Laser beam can be used for the communication between the earth and the moon or to other satellites.
- Laser beam can be used for under water communication, as laser radiation is not absorbed by water.

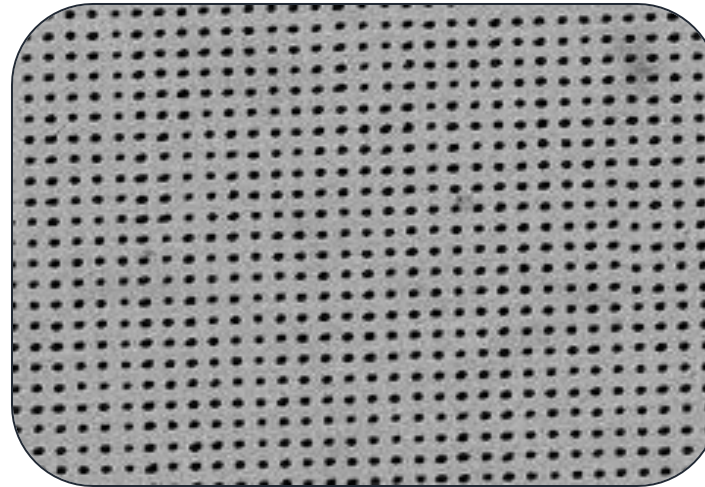


Lasers in Industry:

- Lasers are used for welding. Dissimilar metals can be welded using lasers.
- Holes with controlled precision can be drilled in steel, ceramics, diamond and alloys, using lasers.
- Lasers are widely used in electronic industry in trimming the components of ICs.



Lasers used in welding



Drilling Steel foil for high density filters

Lasers are used in cutting metal sheets, diamond and cloths. In the mass production of stitched clothes, lasers are used to cut the cloth in a desired dimension, all at once.

- Lasers are used for surface treatment. Laser beam is used in selective heat treatment for tempering the desired parts in automobile industry.



Cutting wood using laser



Laser surface treatment to change the micro structure of metals through controlled heating and cooling.



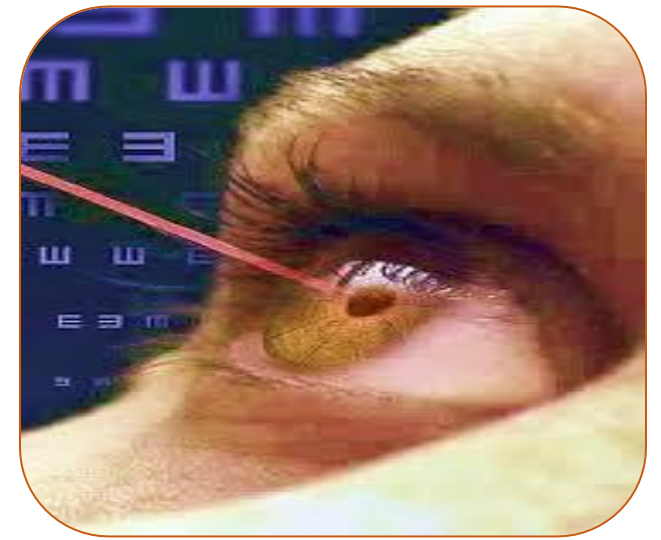
The world's first all-diamond ring, cut with Laser

Lasers in medicine:

- Lasers are used in eye surgery, especially to attach the detached retina.
- Lasers are used for treatments such as plastic surgery, skin injuries and to remove moles, tattoos and tumours developed in skin tissue.
- Lasers are used in stomatology-the study of mouth and its disease.



Lasers used in stomatology



Lasers in Eye surgery



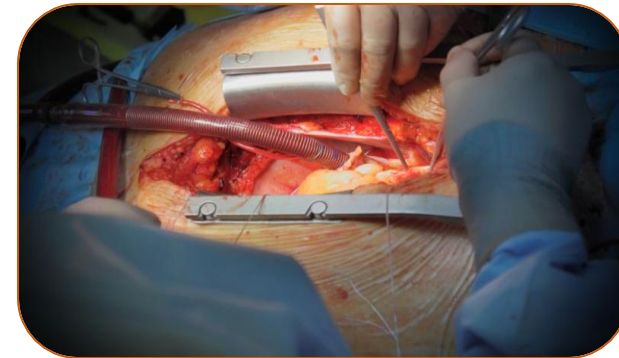
Lasers in tattoo removal

Laser radiation is sent through optical fibre to open the blocked artery region.

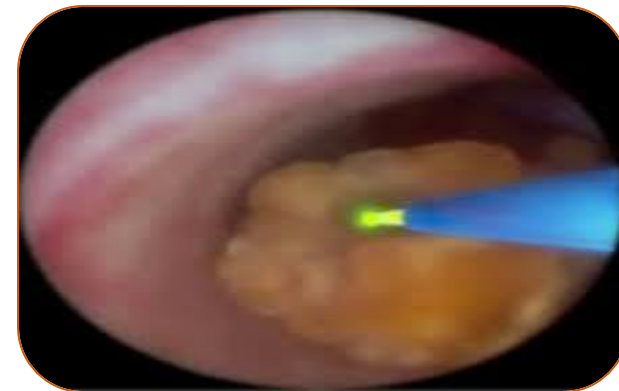
- Lasers are used to destroy kidney stones and gall stones.
- Lasers are used in cancer diagnosis and therapy.
- Lasers are used in blood loss less surgery.
- Lasers are used to control hemorrhage.
- Using CO₂ laser, liver and lung treatment can be carried out.
- Lasers are used in endoscopes, to detect hidden parts.
- Laser Doppler velocimetry is used to measure the velocity of blood in blood vessels.



Red Argon laser used in throat cancer treatment



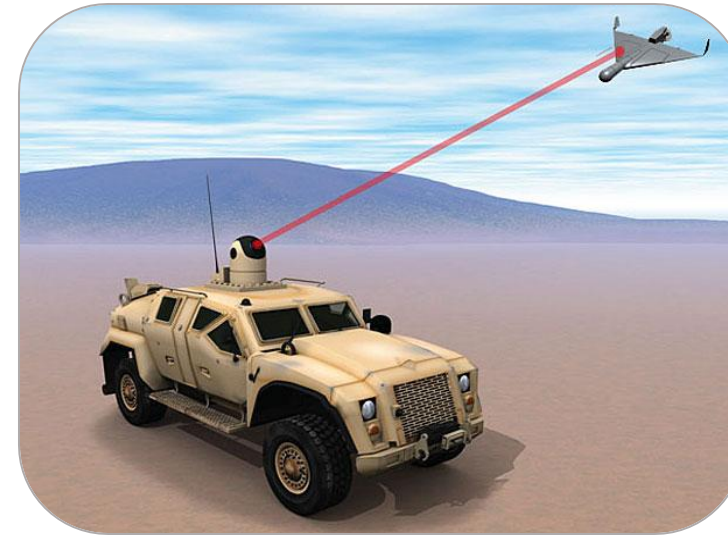
Lasers used to open artery block



Lasers used to destroy kidney stones

Lasers in Military:

- Focusing of high energetic laser beam for few seconds, destroys aircrafts, missiles, etc. These rays are called death rays.
- The vital part of the enemy's body can be evaporated by focusing a highly convergent laser beam from a laser gun.
- LIDAR (Light Detecting And Ranging) is used to estimate the size and shape of distant objects or war weapons.



**Laser armed Humvees shooting a Drone
(flying Robot)**



Lasers beams of RMR LIDAR at ALOMAR Observatory



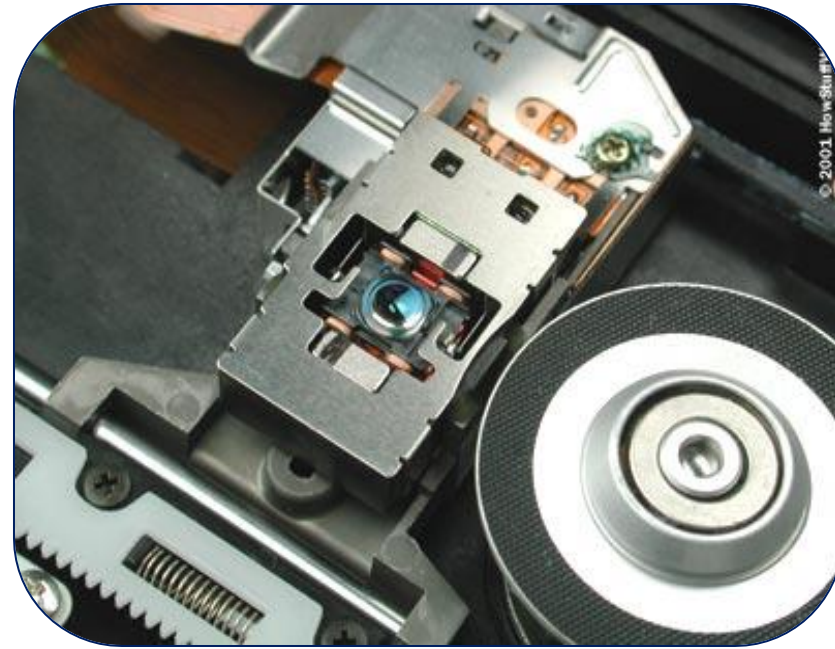
Soldiers using laser gun

Lasers in Computers:

- By using Lasers, a large amount of information or data can be stored in CD-ROM or their storage capacity can be increased.
- Lasers are used in computer printers.



Laser Beam Printer (LBP) by Epson



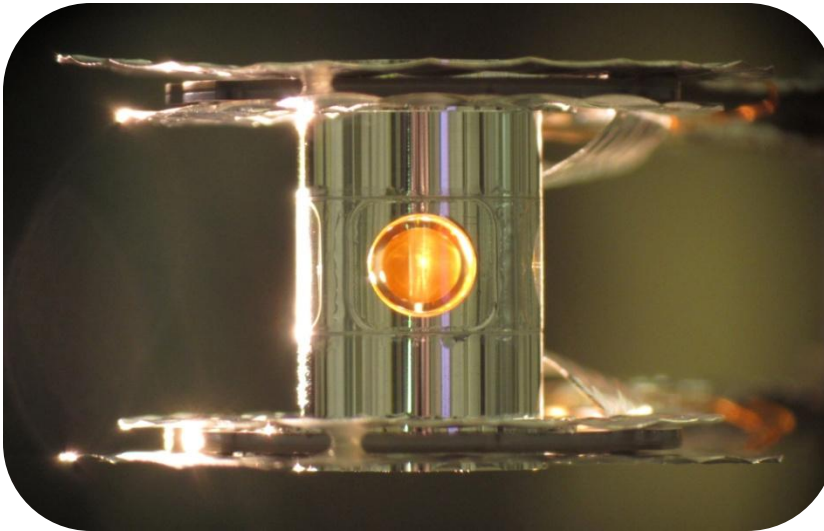
Laser assembly inside a CD burner

Lasers in thermo nuclear fusion:

- A nuclear fusion reaction can be initiated by concentrating a large amount of laser energy in a small volume.
- For example, in the fusion of deuterium and tritium, irradiation with a high energy laser beam, develops a temperature of 10^{17} °C , which is sufficient to initiate nuclear fusion reaction.



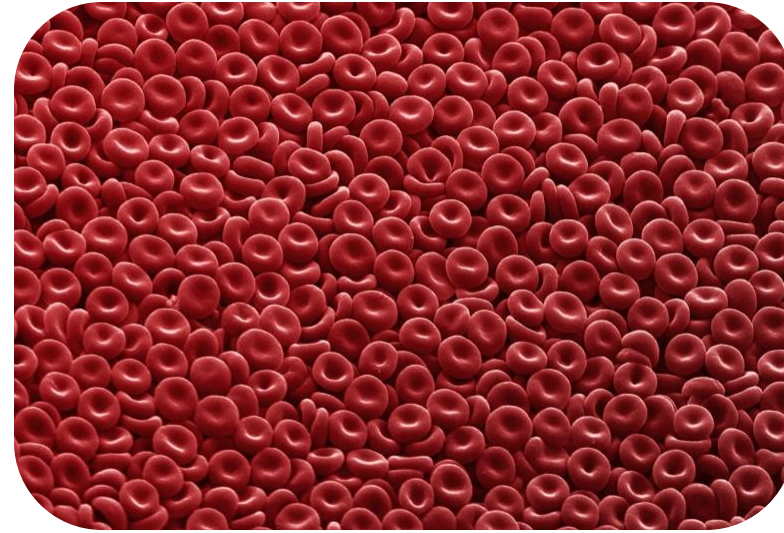
Lasers used in nuclear fusion reactors



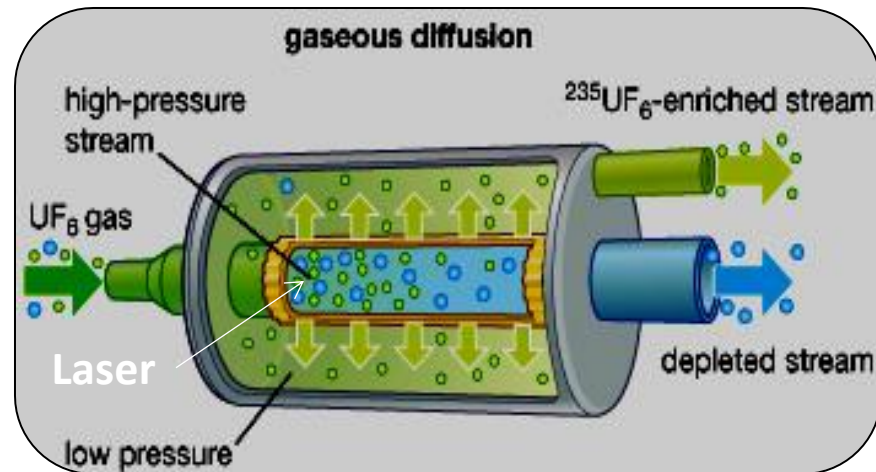
Fusion of deuterium and tritium using 192 lasers

Lasers in Scientific research:

- Laser beam can initiate chemical reactions, study the nature of chemical bonds and also can break molecules.
- Lasers are used to estimate the size and shape of biological cells such as erythrocytes.
- Lasers are used to find the size of dust particles.
- Lasers are used in counting the atoms in isotope separation.

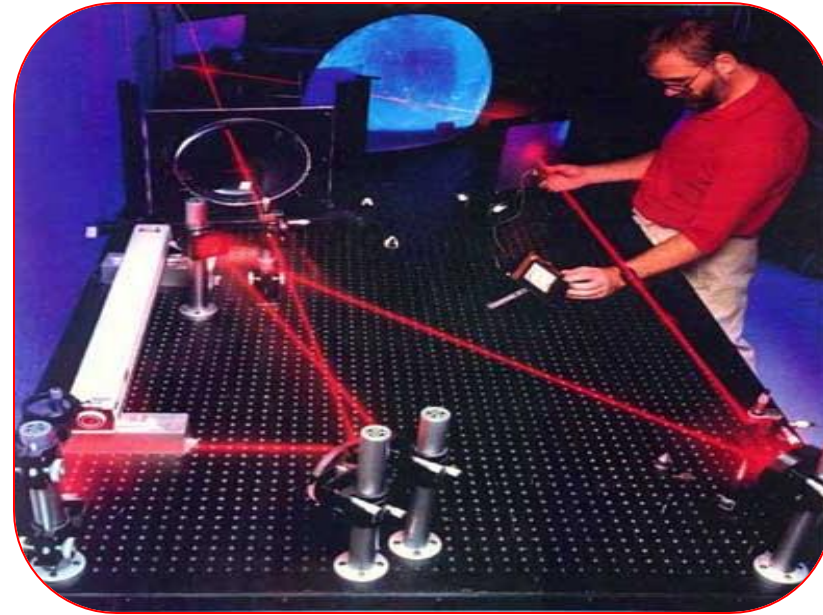


Laser Scanning Microscope micrograph of human RBCs



Laser used in isotope separation

- Lasers are used in holography, for recording and reconstructing of a hologram.
- Lasers are used to measure the constantly changing distance between the moon and the earth, by astronomers.
- Lasers are used in plastic industries to unite monomers to form polymers.
- Lasers are used to develop hidden finger prints and to clean delicate pieces of art.



Lasers used in Holography



Finger print scanning using Laser